EXPERIMENTS 1 TO 20

1.

```
Output

Process ID: 17342
Parent Process ID: 17335
```

3.

```
Output

Enter number of processes: 5
Enter burst time of each process:
P1: 1
P2: 2
P3: 3
P4: 4
P5: 5

Process BT  WT  TAT
P1  1   0   1
P2  2   1   3
P3  3   3   6
P4  4   6   10
P5  5   10  15

Average Waiting Time = 4.00
Average Turnaround Time = 7.00
```

4.

```
Output

Enter number of processes: 1
Enter arrival time and burst time for P1: 1 2

Process AT  BT  WT  TAT
P1  1   2   0   2

Average Waiting Time = 0.00
Average Turnaround Time = 2.00
```

5.

```
Output

Enter number of processes: 3
Enter arrival time, burst time and priority for P1: 1 2 4
Enter arrival time, burst time and priority for P2: 4 5 6
Enter arrival time, burst time and priority for P3: 46 4 6

Process AT  BT  Priority    WT  TAT
P1  1   2   4       0   2
P2  4   5   6       0   5
P3  46  4   6       0   4

Average Waiting Time = 0.00
Average Turnaround Time = 3.67
```

6.

```
Output

Enter number of processes: 3
Enter arrival time, burst time and priority for P1: 0 5 2
Enter arrival time, burst time and priority for P2: 1 3 1
Enter arrival time, burst time and priority for P3: 2 2 3

Process AT  BT  Priority    WT  TAT
P1  0   5   2       3   8
P2  1   3   1       0   3
P3  2   2   3       6   8

Average Waiting Time = 3.00
Average Turnaround Time = 6.33
```

7.

```
Output

Enter number of processes: 3
Enter arrival time and burst time for P1: 4 5
Enter arrival time and burst time for P2: 7 8
Enter arrival time and burst time for P3: 6 3

Process AT  BT  WT  TAT
P1  4   5   0   5
P2  7   8   5   13
P3  6   3   3   6

Average Waiting Time = 2.67
Average Turnaround Time = 8.00
```

8.

Output

```
Enter number of processes: 3
Enter arrival time and burst time for P1: 4 5
Enter arrival time and burst time for P2: 4 6
Enter arrival time and burst time for P3: 5 9
Enter time quantum: 6

Process AT  BT  WT  TAT
P1  4   5   0   5
P2  4   6   5   11
P3  5   9   10  19

Average Waiting Time = 5.00
Average Turnaround Time = 11.67
```

9.

Output

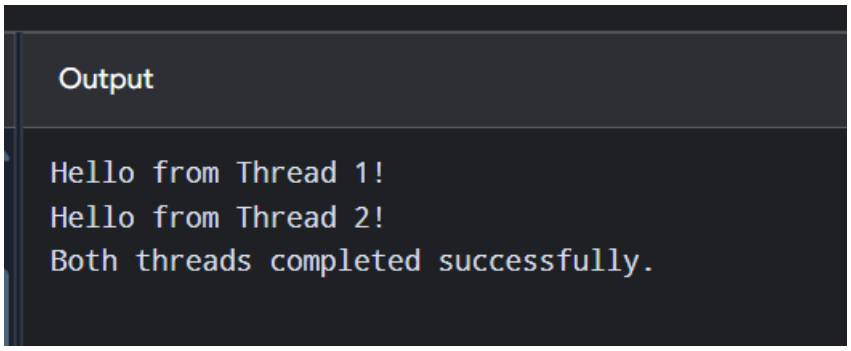
```
Data written to shared memory: Hello, shared memory!
Data read from shared memory: Hello, shared memory!
Shared memory released successfully.
```

10.

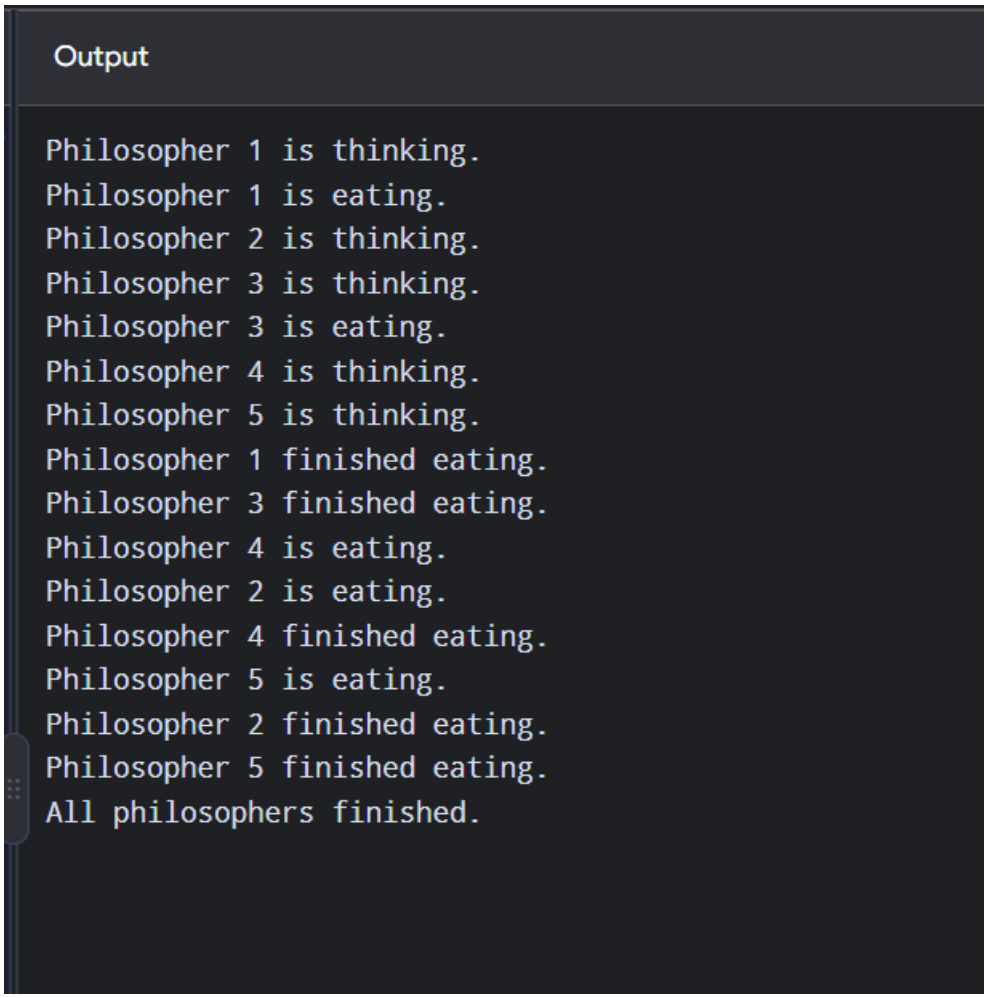
Output

```
Producer: Data sent: Hello, message queue!
Consumer: Data received: Hello, message queue!
Message queue removed successfully.
```

11.


Output
Hello from Thread 1!
Hello from Thread 2!
Both threads completed successfully.

12.


Output
Philosopher 1 is thinking.
Philosopher 1 is eating.
Philosopher 2 is thinking.
Philosopher 3 is thinking.
Philosopher 3 is eating.
Philosopher 4 is thinking.
Philosopher 5 is thinking.
Philosopher 1 finished eating.
Philosopher 3 finished eating.
Philosopher 4 is eating.
Philosopher 2 is eating.
Philosopher 4 finished eating.
Philosopher 5 is eating.
Philosopher 2 finished eating.
Philosopher 5 finished eating.
All philosophers finished.

13.

Output

```
Enter number of memory blocks: 5
Enter sizes of memory blocks:
160
450
500
412
654
Enter number of processes: 4
Enter sizes of processes:
150
252
646
962

Process Size    Block
1   150 1
2   252 2
3   646 5
4   962 Not Allocated
```

14.

Output

```
--- SINGLE LEVEL DIRECTORY ---
1. Create File
2. List Files
3. Search File
4. Delete File
5. Exit
Enter your choice: 1
Enter file name: notes.txt
File created successfully.

--- SINGLE LEVEL DIRECTORY ---
1. Create File
2. List Files
3. Search File
4. Delete File
5. Exit
Enter your choice: 2
Files in directory:
notes.txt
```

15.

```
Output

--- TWO LEVEL DIRECTORY ---
1. Create User Directory
2. Create File
3. List Files
4. Search File
5. Delete File
6. Exit
Enter your choice: 1
Enter user name: user1
User directory created successfully.

--- TWO LEVEL DIRECTORY ---
1. Create User Directory
2. Create File
3. List Files
4. Search File
5. Delete File
6. Exit
Enter your choice: 2
Enter user name: user1
Enter file name: notes.txt
File created successfully.
```

16.

```
Output

Enter number of employees (1-50): 1

Enter details of employee 1
Employee ID: Sreekar
Employee Name (no spaces): Employee Salary: 1234567

Enter record number to access (1-1): 1

--- Employee Details ---
Employee ID: 155648
Employee Name: Sreekar
Employee Salary: 1234567.00
3
```

17.

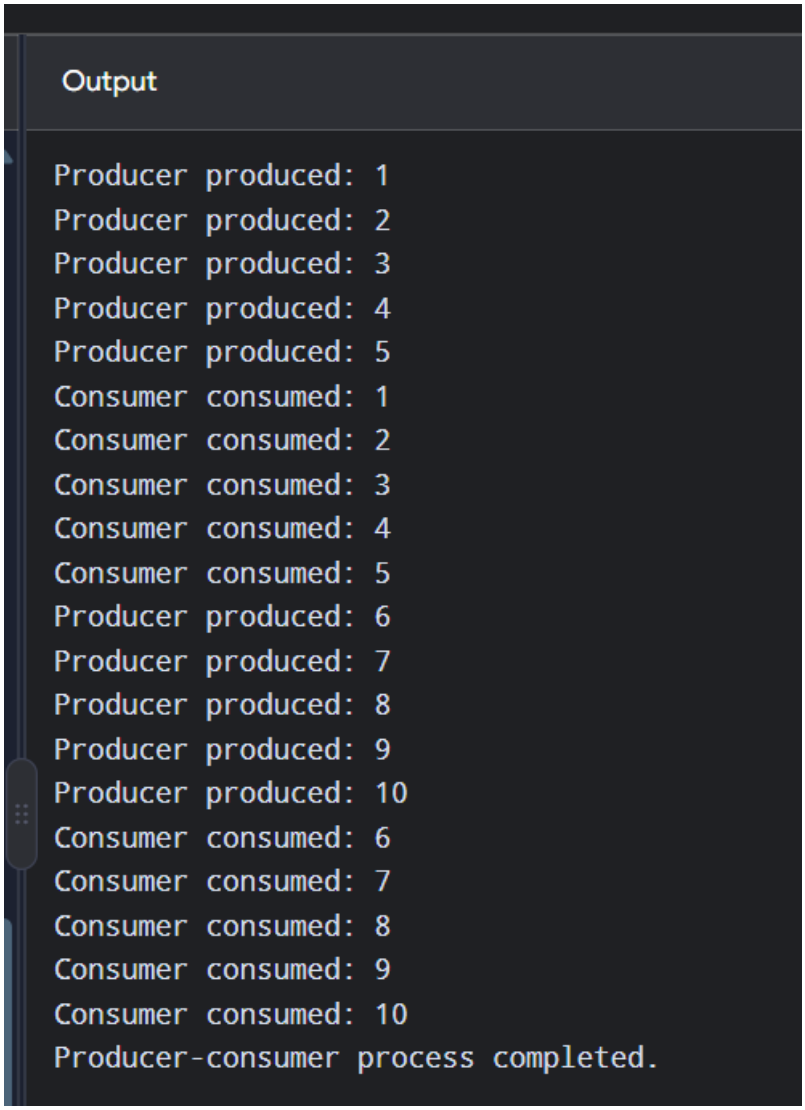
Output

```
Enter number of processes: 5
3
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2
7 5 3
3 2 2
9 0 2
2 2 2
4 3 3
3 3 2
Enter number of resource types: Enter Allocation Matrix:
Enter Maximum Matrix:
Enter Available Resources:

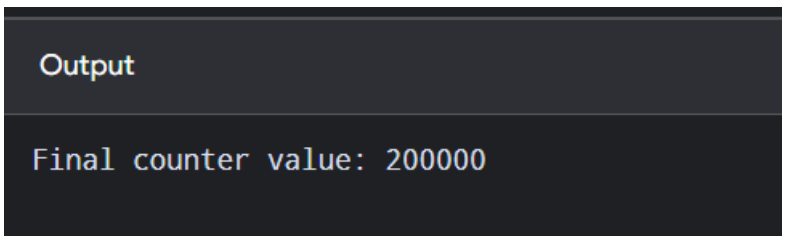
Need Matrix:
7 4 3
1 2 2
6 0 0
0 1 1
4 3 1

System is in a SAFE state.
Safe sequence: P1 -> P3 -> P4 -> P0 -> P2
```

18.


Output
Producer produced: 1
Producer produced: 2
Producer produced: 3
Producer produced: 4
Producer produced: 5
Consumer consumed: 1
Consumer consumed: 2
Consumer consumed: 3
Consumer consumed: 4
Consumer consumed: 5
Producer produced: 6
Producer produced: 7
Producer produced: 8
Producer produced: 9
Producer produced: 10
Consumer consumed: 6
Consumer consumed: 7
Consumer consumed: 8
Consumer consumed: 9
Consumer consumed: 10
Producer-consumer process completed.

19.

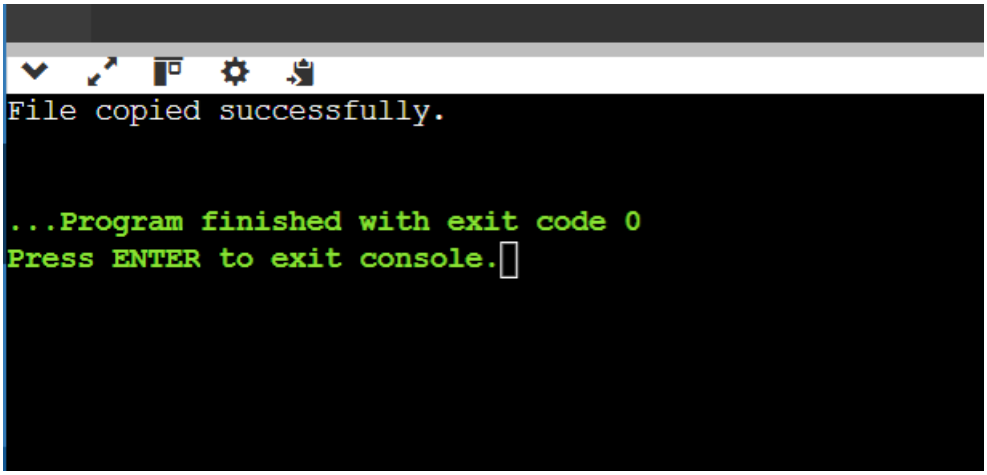

Output
Final counter value: 200000

20.

```
Output

Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Reader reads data: 0
Writer writes data: 1
Writer writes data: 2
Writer writes data: 3
Writer writes data: 4
Writer writes data: 5
Writer writes data: 6
Writer writes data: 7
Writer writes data: 8
Writer writes data: 9
Writer writes data: 10
Reader-writer problem completed.
```

2.



21.

Output

```
Enter number of memory blocks: 3
Enter size of each memory block:
200
345
466
Enter number of processes: 3
Enter size of each process:
232
121
465

Process Process Size    Block Allocated
1   232     3
2   121     2
3   465     Not Allocated
```

22.

```
Enter number of memory blocks: 4
Enter size of each memory block:
564
465
646
867
Enter number of processes: 3
Enter size of each process:
765
455
865

Process Size    Block Allocated
P1  765 4
P2  455 2
P3  865 Not Allocated
```

23.

```
Output

Enter number of memory blocks: 2
Enter size of each memory block:
234
566
Enter number of processes: 2
Enter size of each process:
4566
543

Process Size    Block Allocated
P1  4566    Not Allocated
P2  543 2
```

24.

```
Data written successfully.
File contents: UNIX File Management System Calls
File closed successfully.


...Program finished with exit code 0
Press ENTER to exit console.
```